

A Comparative Study of DC-DC Converters for Maximum Power Point Tracking of Solar Array

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Abstract—In recent decades, climate change has forced the energy sector to shift towards renewable sources of energy. This change has highlighted the importance of solar photovoltaic (PV) systems for sustainable power generation. In PV systems, a Maximum Power Point Tracking (MPPT) technique must be implemented to continuously track the Maximum Power Point of the PV array to maximize efficiency. A comparison of five DC-DC converter topologies—Boost, Buck-Boost, Cuk, SEPIC, and Zeta—for MPPT of a solar array is presented in this study. The study evaluates the efficiency, response time, and output ripple of each converter under varying irradiance and temperature conditions. The simulation results obtained using MATLAB/Simulink demonstrate the strengths and limitations of each topology. The findings aim to guide the selection of optimal DC-DC converters to improve the overall efficiency of PV systems.

Index Terms—MPPT, PID Controller, DC-DC converter, Solar power, MATLAB/Simulink

I. INTRODUCTION

The world is moving toward renewable energy sources as a key strategy to address climate change and decrease dependence on fossil fuels. Among these, solar energy stands out as a clean, sustainable, and widely available resource. Countries such as Bangladesh, with their tropical climate and abundant sunlight, are ideal positioned to harness solar power. On average, Bangladesh receives solar radiation of 4 to 6.5 kWh per square meter per day in most of the country, which represents significant potential for solar energy generation [1]. However, despite the growing interest in solar energy, large-scale implementation remains underdeveloped, with much of the country's potential still untapped. However, photovoltaic systems (PV) are at the heart of solar energy conversion, consisting of photovoltaic arrays that convert sunlight into electricity. The drawback of these systems is that they suffer from a non-linear nature [2]. Due to the fact that solar irradiance and temperature conditions vary throughout a day, PV systems require a DC-DC converter to ensure efficient power extraction. A key challenge in photovoltaic systems is operating the array at its Maximum Power Point (MPP), where the energy harvested is maximized. To address this, Maximum Power Point Tracking (MPPT) algorithms are employed which adjust the duty cycle of the converters [3]. Among the widely used MPPT techniques, the PID (Proportional-Integral-Derivative) control algorithm offers

precise and reliable control, making it a suitable choice for PV systems [4].

Several studies have been done in MPPT based Solar Arrays [5] [6] [7]. However, these studies examine only one or two converter topologies, providing limited insights into selecting the most effective converter for MPPT in PV systems. Rastogi et al. conducted a comparative study on five converter topologies for MPPT of the solar array [8]. The performance metrics involved in the paper were only the efficiency and cost of the converters. Other important parameters such as output ripple and response time of the converters have not been shown. A comparison of six different DC-DC converters is reported in [9]. In this paper, the analysis of each converter's operational characteristics under varying radiation and temperature conditions is carried out using both theoretical analysis and simulation results. An analysis of two different MPPT algorithms for five DC-DC converter topologies is shown in [10] but tracking speed and efficiency are the only performance metrics reported. Mohammad et al. conducted a study on three different DC-DC converters for MPPT of PV array [11]. The study tested two different MPPT algorithms on boost, buck-boost and Cuk converter to find efficiency, voltage ripple and current ripple. The drawback of this paper is that the temperature and irradiance conditions were not varied.

This paper presents a comparison of five different DC-DC converter topologies: Boost, Buck-Boost, Cuk, SEPIC, and Zeta.. While various MPPT methods exist, a perturb and observe algorithm is employed with a PID controller to ensure stable and accurate tracking of the maximum power point. The performance metrics compared in this study are: efficiency, settling time, voltage ripple and current ripple.

II. SYSTEM MODEL

The photovoltaic solar system consists of a solar array, a DC-DC converter, and an MPPT controller. The converter receives voltage and current as input from the photovoltaic array and connects to a load. The MPPT controller monitors the changing voltage and current and its algorithm processes these inputs to optimize performance. The block diagram of the system is illustrated in Fig. 1.

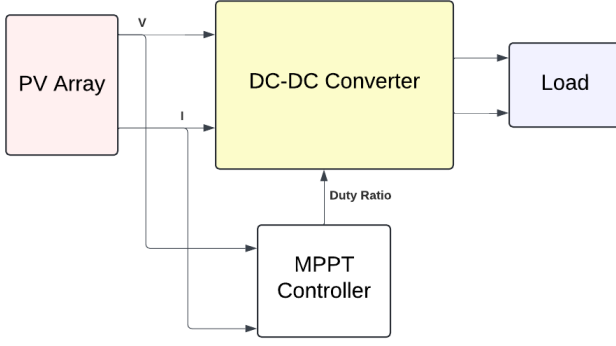


Fig. 1: Layout of PV System.

A. PV Array

The PV array is the main transducer in the study. It receives irradiance and temperature as input and gives voltage and current as output in the system. A10 Green Technology A10J-S72-175 module was selected for the study. The standard value of irradiance and temperature was chosen as 1000 W/m^2 and 25°C . PV array usually consists of a number of series and parallel connected solar cells. However, for simplicity, a single series and parallel cell was chosen for this study. Table I displays the properties of the module.

TABLE I: PV module Parameter values

Parameter	Symbol	Value
Cells per module	N _{cell}	72
Maximum power	$P_{\max}$	175 W
Open circuit voltage	V_{oc}	43.99 V
Short circuit current	I_{sc}	5.17 A
Temperature coefficient of current	K_i	$0.041509 \text{ A/}^\circ\text{C}$
Temperature coefficient of voltage	K_v	$-0.3616 \text{ mV/}^\circ\text{C}$
Current at maximum power	$I_{\max}$	4.78 A
Voltage at maximum power	$V_{\max}$	36.63 V
Light-generated current	I_L	5.178 A
Diode saturation current	I_0	$1.7842 \times 10^{-10} \text{ A}$
Ideality factor	A	0.98852
Shunt resistance	R_{sc}	249.7Ω
Series resistance	R_s	0.38412Ω

B. MPPT Controller

A MPPT controller was implemented using a perturb and observe algorithm and a PID controller to optimize the power output of the photovoltaic system. Fig. 2 exhibits the flow chart of the perturb and observe algorithm [12]. The voltage and current of the solar array was fed into the MPPT algorithm block to generate the reference voltage. An error signal is then created by calculating the difference between PV array's output voltage and the reference voltage, which is then fed into the PID controller. The output of the PID controller gives the duty cycle which is sent to a PWM generator for PWM signal generation. This PWM signal is sent to the MOSFET's gate for controlling the converter circuit's switching. The Simulink model of the MPPT subsystem is shown in Fig. 3.

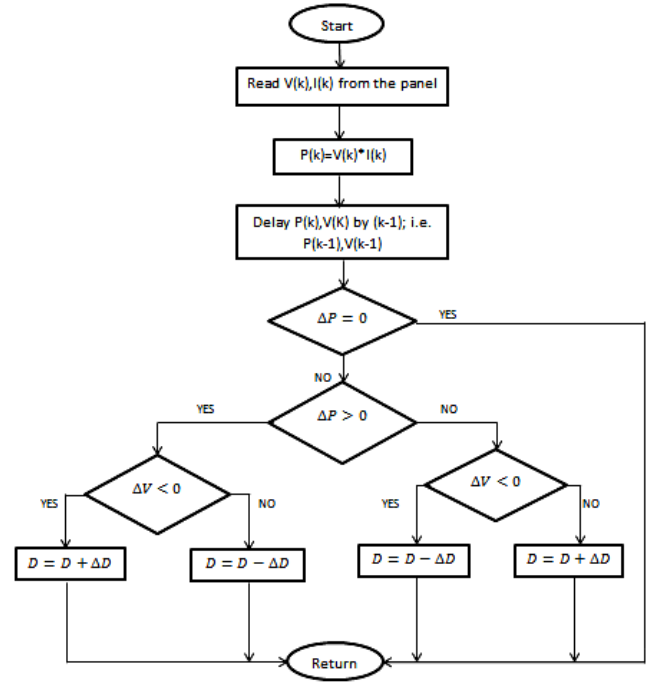


Fig. 2: Flowchart of Perturb and Observe Algorithm.

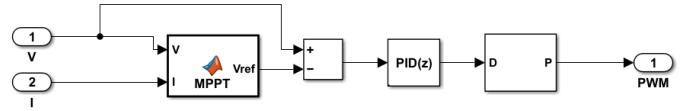


Fig. 3: Simulink Model of MPPT Controller.

C. DC-DC Converters

Solar arrays produce DC power but their voltage generated can vary due to fluctuating environmental conditions. Each array has a maximum power point where it can produce the most power. By controlling the duty cycle of a switching device, such as a MOSFET, DC-DC converters facilitate consistent operation at the maximum power point (MPP), effectively accommodating fluctuations in temperature and solar irradiance. This paper analyzes five key DC-DC converter topologies—Boost, Cuk, SEPIC, Buck-Boost and Zeta converters—emphasizing their roles in MPPT and voltage stabilization in PV systems.

1) *Boost Converter*: The boost converter generates an output voltage that is larger than the input voltage. There is a direct correlation between the duty cycle (D) and the output voltage; as D rises, the output voltage also increases. The MOSFET handles the switching process, while inductors and capacitors are used for energy storage and filtering. When the switch is closed, the diode becomes reverse biased, so energy is stored by the inductor. When the switch becomes open,

the diode becomes forward biased which enables the stored energy to transfer to the load. Additionally, the capacitor is essential for smoothing the output voltage, promoting stable performance. The schematic diagram of the boost converter is displayed in Fig. 4.

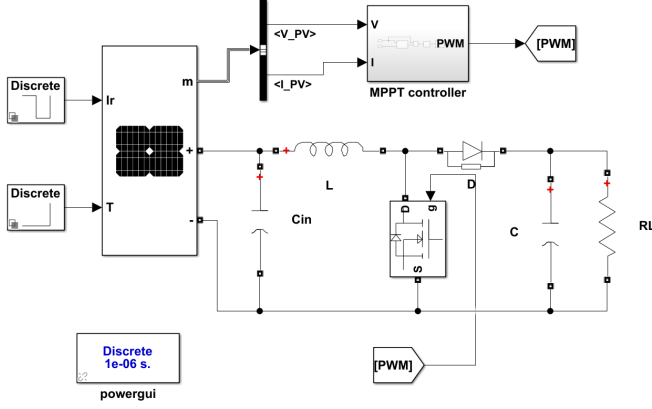


Fig. 4: Schematic Diagram of DC-DC Boost Converter.

2) *Buck-Boost Converter*: The buck-boost converter can step up or down the input voltage as required by the system. The condition for the converter to step-up or step-down voltage depends on the duty cycle of the switch. When the duty cycle is above 0.5, the converter acts as a boost converter, raising the output voltage. However, if the duty cycle is below 0.5, it reduces the output voltage. This dual capability makes the buck-boost converter especially useful for applications such as MPPT in photovoltaic systems, where fluctuating input conditions demand flexible voltage regulation. The schematic diagram of the buck-boost converter is displayed in Fig. 5.

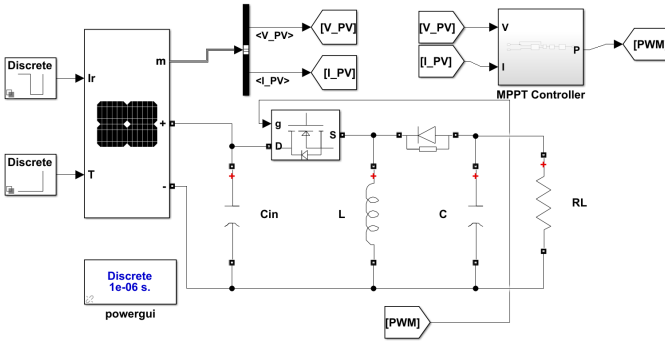


Fig. 5: Schematic Diagram of DC-DC Buck-Boost Converter.

3) *Cuk Converter*: This converter can also increase or decrease the output input voltage. It uses a single switching device, minimizing switching losses. A key feature of the Cuk converter is an extra capacitor for which stores energy and an inductor at the output, which improves current stability. During the switch's off-time, the inductor current charges the capacitor via a diode, and during the on-time, the diode is reverse biased, allowing current to pass via the switch and maintain efficient performance. The schematic diagram of the Cuk converter is shown in Fig. 6.

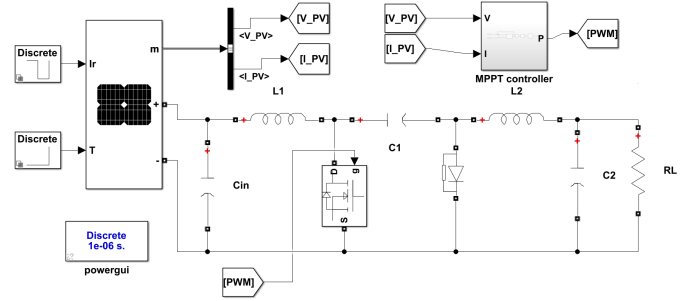


Fig. 6: Schematic Diagram of DC-DC Cuk Converter.

4) *SEPIC Converter*: The SEPIC converter is an upgraded version of the buck-boost and Cuk converter in the sense that they produce a non-inverting output. In the SEPIC setup, current travels to the first inductor when the switch is closed, while simultaneously charging the second inductor via the capacitor. When the switch is open, the inductors release energy to the load through a diode. A notable benefit of the SEPIC converter is its ability to reduce voltage stress on the capacitor, thereby increasing its lifespan and reliability compared to the Cuk converter. The schematic diagram of the SEPIC converter is presented in Fig. 7.

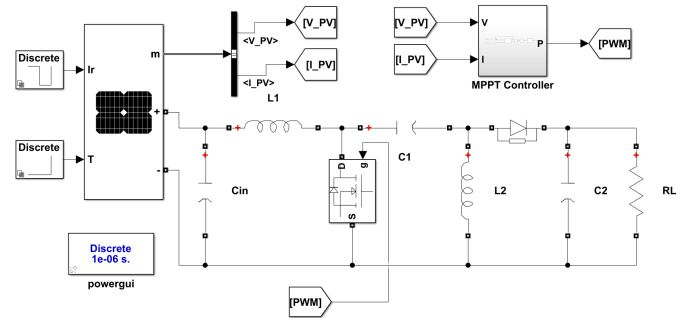


Fig. 7: Schematic Diagram of DC-DC SEPIC Converter.

5) *Zeta Converter*: The Zeta converter, like the SEPIC, uses a two-inductor topology, allowing it to both increase and decrease voltage while providing a non-inverting output voltage. When the switch is closed, the first inductor ($L1$) stores energy, which is then transferred to the capacitor ($C1$) when the diode is forward-biased. In the next cycle, the energy in $C1$ flows to the load via the second inductor ($L2$), ensuring a smooth and stable output voltage. The schematic diagram of the Zeta converter is shown in Fig. 8.

III. RESULTS AND ANALYSIS

The performance of the five converters was evaluated through simulations conducted in MATLAB/Simulink. The values of capacitance and inductance of each converter were determined through a trial and error method. The initial values chosen for the circuit parameters were high to reduce ripple. The parameter values were then updated based on the smoothness of the curves and response time. A DC link capacitor has been added to each converter circuit to reduce

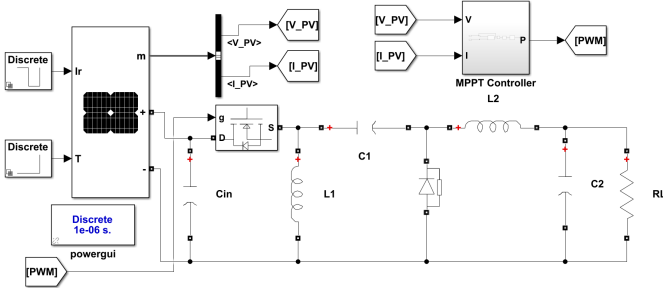


Fig. 8: Schematic Diagram of DC-DC Zeta Converter.

output ripple. The load has been fixed at 125Ω for control. In order to simulate the variable environmental conditions of the real world, the simulation was done under three separate conditions:

- 1) Standard conditions: Irradiance at 1000 W/m^2 and temperature at 25°C .
- 2) Irradiance change: Irradiance at 500 W/m^2 and temperature at 25°C .
- 3) Temperature change: Irradiance at 1000 W/m^2 and temperature at 50°C .

Fig. 9 to Fig. 13 show the input and output power of the DC to DC converters. The maximum input power(P_{in}) and output power(P_{out}) of all the DC-DC converters were recorded and the efficiency of the converters were calculated using the formula:

$$\eta = \frac{P_{in}}{P_{out}} \times 100 \quad (1)$$

Furthermore, the response time for each converter to reach its maximum power was observed and noted down for all three simulation scenarios. The voltage ripple and current ripple of the DC-DC converters were computed by the following formula:

$$X = \frac{X_{max} - X_{min}}{X_{avg}} \times 100 \quad (2)$$

where,

X: Current or Voltage

X_{max} : Maximum value of current or voltage

X_{min} : Minimum value of current or voltage

X_{avg} : Average value of current or voltage

Table II gives a comprehensive summary of the simulation results. The boost converter achieved the highest efficiency at 95.9%, however it exhibited high voltage and current ripple of almost 1% and also showed the longest settling time in all three simulation periods. The other converters showed similar efficiency levels-ranging from 92.8% to 93.5%. They also had comparable settling times. The Zeta converter had the lowest ripple at just 0.09% while the SEPIC converter had the highest ripple at 1.29%.

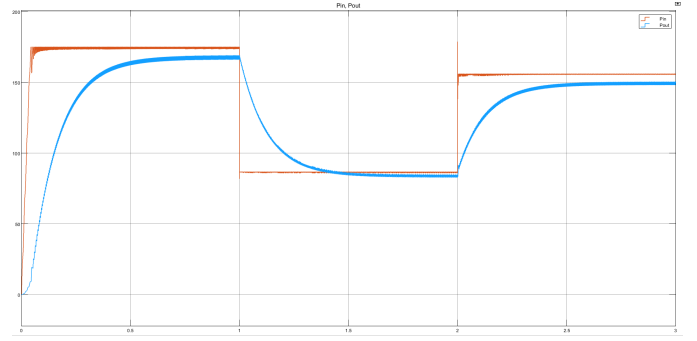


Fig. 9: Response of Boost Converter.

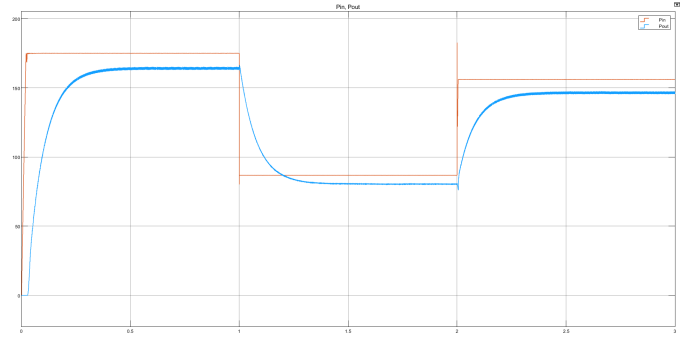


Fig. 10: Response of Buck-Boost Converter.

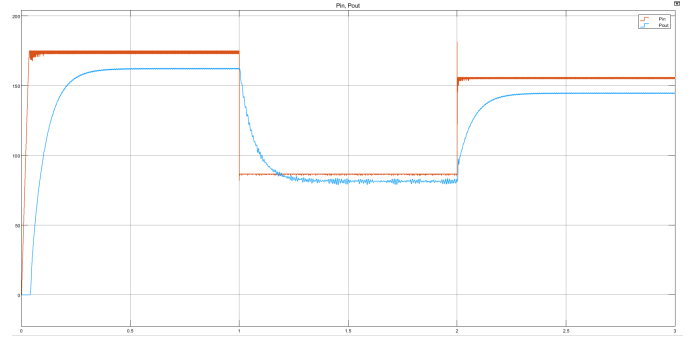


Fig. 11: Response of Cuk Converter.

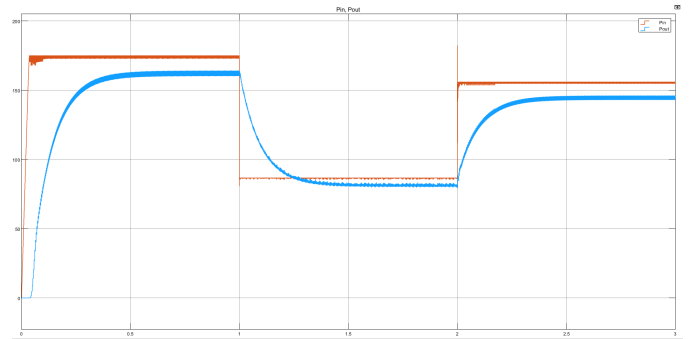


Fig. 12: Response of SEPIC Converter.

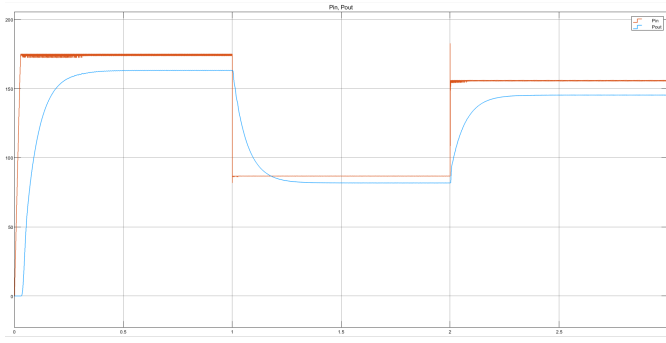


Fig. 13: Response of Zeta Converter.

TABLE II: Performance Metrics of DC-DC Converters

Parameters	Boost	Buck-boost	Cuk	Sepic	Zeta
Pin(W)	174.5	174.5	174.5	174.5	174.5
Pout(W)	167.3	163.7	162.2	161.9	163.2
Efficiency(%)	95.9	93.5	93.0	92.8	93.5
Voltage Ripple(%)	0.97	0.66	0.29	1.23	0.09
Current Ripple(%)	0.86	0.65	0.29	1.23	0.09
Settling time in first third(s)	0.80	0.50	0.46	0.65	0.47
Settling time in second third(s)	1.85	1.50	1.38	1.59	1.42
Settling time in final third(s)	2.68	2.50	2.36	2.60	2.37

IV. CONCLUSION

In this paper, the Maximum Power Point of a photovoltaic (PV) power system was tracked using five different DC-DC converters with a PID controller. Our analysis revealed that the Boost converter-despite its high efficiency, is best suited for stable environments due to its sensitivity to varying environmental conditions. The Buck-Boost, Cuk and Zeta converters provide a good balance between efficiency and stability. From these results, it can be said that in regions where the irradiance and temperature change constantly, the buck-boost, cuk or zeta converter would be the optimum choice. The SEPIC converter displayed the worst performance out of all five converters.

There is scope for further research on this topic. Additional weather data such as cloud cover and seasonal variations can

be incorporated to assess the performance of these converters over a long period of time.

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